Allergic Rhinitis

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Allergic rhinitis is seasonal or perennial itching, sneezing, rhinorrhea, nasal congestion, and sometimes conjunctivitis, caused by exposure to pollens or other allergens. Diagnosis is by history and occasionally skin testing. First-line treatment is with a nasal corticosteroid (with or without an oral or a nasal antihistamine) or with an oral antihistamine plus an oral decongestant.

(See also Overview of Allergic and Atopic Disorders.)

Allergic rhinitis may occur seasonally or throughout the year (as a form of perennial rhinitis). Seasonal rhinitis is usually allergic. At least 25% of perennial rhinitis is nonallergic.

Seasonal allergic rhinitis (hay fever) is most often caused by plant allergens, which vary by season and geographic location. Common plant allergens include

- **Spring:** Tree pollens (eg, oak, elm, maple, alder, birch, juniper, olive)

- **Summer:** Grass pollens (eg, Bermuda, timothy, sweet vernal, orchard, Johnson) and weed pollens (eg, Russian thistle, English plantain)
- **Fall:** Other weed pollens (eg, ragweed)

Causes also differ by region, and seasonal allergic rhinitis is occasionally caused by airborne fungal (mold) spores.

Perennial rhinitis is caused by year-round exposure to indoor inhaled allergens (eg, dust mite feces, cockroach components, animal dander) or by strong reactivity to plant pollens in sequential seasons.

Allergic rhinitis and asthma frequently coexist; whether rhinitis and asthma result from the same allergic process (one-airway hypothesis) or rhinitis is a discrete asthma trigger is unclear.

The numerous nonallergic forms of perennial rhinitis include infectious, vasomotor, medication-induced (eg, aspirin- or nonsteroidal anti-inflammatory drug [NSAID]–induced), and atrophic rhinitis and nonallergic rhinitis with eosinophilia (NARES).

Symptoms and Signs of Allergic Rhinitis

Patients have itching (in the nose, eyes, or mouth), sneezing, rhinorrhea, and nasal and sinus obstruction. Sinus obstruction may cause frontal headaches; viral or bacterial sinusitis is a frequent complication of allergic rhinitis. Coughing and wheezing may also occur, especially if asthma is also present.

The most prominent feature of perennial rhinitis is chronic nasal obstruction, which, in children, can lead to chronic otitis media; symptoms vary in severity throughout the year. Itching is less prominent than in seasonal rhinitis. Chronic sinusitis and nasal polyps may develop.

Signs include edematous, bluish-red nasal turbinates and, in some cases of seasonal allergic rhinitis, conjunctival injection and eyelid edema.

If allergic conjunctivitis is present, symptoms include bilateral mild to severe ocular itching, conjunctival hyperemia, photosensitivity, eyelid edema, and a watery or stringy discharge.

Diagnosis of Allergic Rhinitis

- History and physical examination

- Occasionally skin testing, allergen-specific serum IgE tests, or both

Allergic rhinitis can frequently be diagnosed based on history alone. Diagnostic testing is not routinely needed unless patients do not improve when treated empirically; for such patients, skin tests are done to identify a reaction to pollens (seasonal) or to dust mite feces, cockroaches, animal dander, mold, or other antigens (perennial), which can be used to guide additional treatment.

Occasionally, skin test results are equivocal, or testing cannot be done (eg, because patients are taking medications that interfere with results and cannot safely be withheld); then, an allergen-specific serum IgE test is done.

A nasal smear to detect eosinophils can be done to confirm allergic rhinitis. Although the test is rarely done, it can be a useful alternative to needlesticks in children or used as an additional tool in the evaluation of rhinitis. The sensitivity and specificity of the test is not clear (1). Eosinophilia detected on nasal smear plus negative skin tests suggests aspirin sensitivity or nonallergic rhinitis with eosinophilia (NARES).

Nonallergic perennial rhinitis is usually also diagnosed based on history. Lack of a clinical response to treatment for assumed allergic rhinitis and negative results on skin tests and/or an allergen-specific serum IgE test also suggest a nonallergic cause; disorders to consider include nasal tumors, enlarged adenoids, hypertrophic nasal turbinates, granulomatosis with polyangiitis, and sarcoidosis.

Diagnosis reference

1. Pal I, Babu AS, Halder I, Kumar S: Nasal smear eosinophils and allergic rhinitis. *Ear Nose Throat* 96 (10-11):E17–E22, 2017. doi: 10.1177/0145561317096010-1105

Treatment of Allergic Rhinitis

- Antihistamines
- Decongestants
- Nasal corticosteroids
- For seasonal or severe refractory rhinitis, sometimes desensitization

Treatment of seasonal and perennial allergic rhinitis is generally the same, although attempts at removal or avoidance of allergens (eg, eliminating dust mites and cockroaches) are recommended for perennial rhinitis. For seasonal or severe refractory rhinitis, desensitization immunotherapy may help.

The most effective first-line medication treatments are

- Intranasal corticosteroids with or without oral or intranasal antihistamines
- Oral antihistamines plus oral decongestants (eg, a sympathomimetic such as pseudoephedrine)

Intranasal decongestant sprays (eg, oxymetazoline, phenylephrine) are used for short-term relief of nasal congestion. Usually, they are not recommended for use for more than 3 consecutive days because rebound nasal congestion may occur.

Less effective alternatives include nasal mast cell stabilizers (eg, cromolyn), the intranasal H1 blocker azelastine, and nasal ipratropium which relieves rhinorrhea.

A combination of azelastine/fluticasone (137 mcg/50 mcg) is available.

Intranasal medications are often preferred to oral medications because less of the medication is absorbed systemically.

Intranasal saline, often forgotten, helps mobilize thick nasal secretions and hydrate nasal mucous membranes; various saline solution kits and irrigation devices (eg, squeeze bottles, bulb syringes) and kits are available over the counter, or patients can make their own solutions.

Montelukast, a leukotriene inhibitor, relieves allergic rhinitis symptoms but, due to a risk of psychiatric adverse effects (eg, hallucination, obsessive-compulsive disorder, suicidal thoughts and behavior), montelukast should be used only when other treatments are not effective or not tolerated.

Omalizumab, an anti-IgE antibody has been used to treat allergic rhinitis but probably has a limited role because less expensive, effective alternatives are available.

Prevention of Allergic Rhinitis

- Avoidance of triggers
- Sometimes desensitization

Avoidance of triggers

For perennial allergies, triggers should be removed or avoided if possible. Strategies include the following:

- Removing items that collect dust, such as knickknacks, magazines, books, and soft toys
- Using synthetic fiber pillows and impermeable mattress covers
- Frequently washing bed sheets, pillowcases, and blankets in hot water
- Frequently cleaning the house, including dusting, vacuuming, and wet-mopping
- Removing upholstered furniture and carpets or frequently vacuuming them
- Replacing draperies and shades with blinds
- Exterminating cockroaches to eliminate exposure
- Using dehumidifiers in basements and other poorly aerated, damp rooms
- Using high-efficiency particulate air (HEPA) vacuums and filters
- Avoiding food or medication triggers
- Limiting pets to certain rooms or keeping them out of the house
- For people with severe seasonal allergies, possibly moving to an area that does not have the allergen

Adjunctive nonallergenic triggers (eg, cigarette smoke, strong odors, irritating fumes, air pollution, cold temperatures, high humidity) should also be avoided or controlled when possible.

Desensitization

Desensitization immunotherapy may be more effective for seasonal than for perennial allergic rhinitis; it is indicated when

- Symptoms are severe.
- The allergen cannot be avoided.
- Medication treatment is inadequate.

First attempts at desensitization should begin soon after the pollen season ends to prepare for the next season; adverse reactions increase when desensitization is started during the pollen season because the person's allergic immunity is already maximally stimulated.

Sublingual immunotherapy using 5–grass pollen sublingual tablets can be used to treat grass pollen–induced allergic rhinitis. An alternative is a sublingual tablet of timothy grass extract

The first dose is given in a health care setting. *Patients should be observed for 30 minutes after administration because anaphylaxis may occur*. If the first dose is tolerated, patients can take subsequent doses at home. Treatment is initiated 4 months before the onset of each grass pollen season and maintained throughout the season.

Sublingual immunotherapy using either ragweed pollen or house dust mite allergen extracts can be used to treat allergic rhinitis induced by these allergens.

Patients taking home immunotherapy for allergic rhinitis should carry a prefilled, self-injecting epinephrine syringe to use in case of an anaphylactic reaction.

Key Points

- **Seasonal rhinitis is usually an allergic reaction to pollens; perennial allergic rhinitis is caused by year-round exposure to indoor inhaled allergens or by strong reactivity to plant pollens in sequential seasons.**
- **Patients with allergic rhinitis may have cough, wheezing, frontal headache, sinusitis, or, particularly in children with perennial rhinitis, otitis media.**
- **Diagnosis of allergic rhinitis is usually based on the history; skin tests and sometimes an allergen-specific serum IgE test are needed only when patients do not respond to empiric treatment.**
- **Try intranasal corticosteroids first because they are the most effective treatment and have few systemic effects.**
- **Other treatments include oral and intranasal antihistamines and decongestants; intranasal mast cell stabilizers are less effective.**
- **Desensitization sublingual immunotherapy is indicated when symptoms are severe, allergens cannot be avoided, or medication treatment is inadequate.**

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